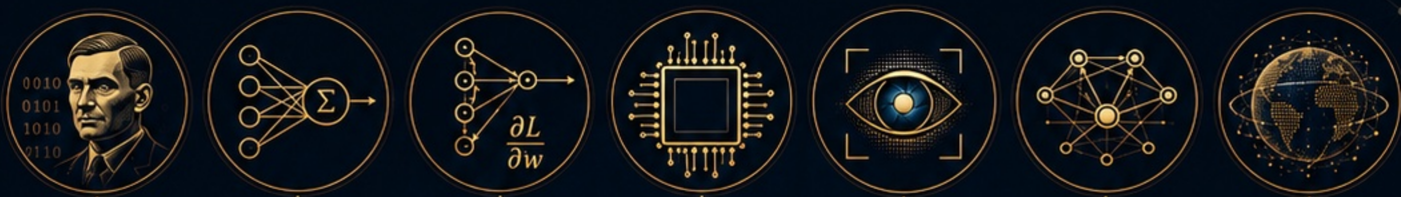
Attention is All You Need

A Brief History of AI & Neural Networks

From symbolic dreams to foundation models



- **1940s–1950s** — The Dream Begins
- **1960s–1970s** — First Disappointment
- **1980s** — Backpropagation
- **1990s–2000s** — Slow Burn
- **2012** — Deep Learning Breakthrough
- **2017** — Transformer Arrives
- **2020s** — Foundation Models

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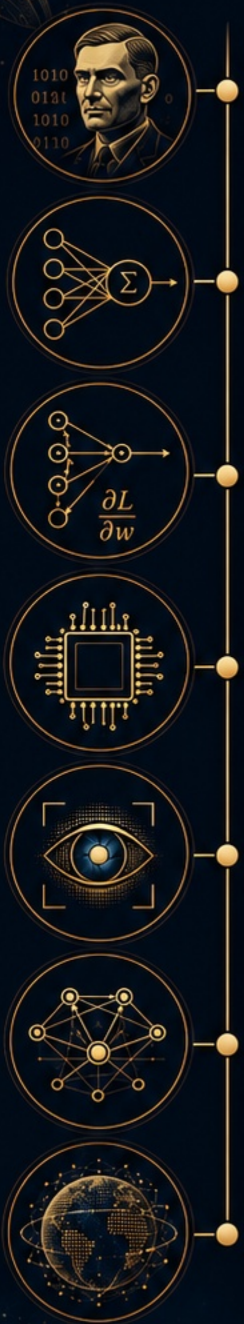

INTRODUCTION

Once upon a time, humans looked at computers which used to be giant, glorified calculators that filled entire rooms and thought:

But what if it could dream?

We tend to look at today's AI chatbots, image generators, and coding assistants as if they dropped out of a sci-fi movie last Tuesday. But the truth is much funnier, weirder, and more human. Modern AI isn't a sudden miracle; it's an 80-year-old math project that went through several existential crises, a few bankruptcies, and a lot of awkward teenage phases.

Grab some popcorn. This is the biography of an idea that refused to die, moving through a spiral of

Dreams, Errors, Patience, Sight, Attention, and Scale.

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Act I: The Dreamer in the Room (1940s–1950s)

Our story kicks off in a world of jazz, black-and-white television, and a British math genius named **Alan Turing.** Turing looked at the massive, clunky computers of his day and decided to set a philosophical trapdoor. He proposed the **"Imitation Game" (The Turing Test):** Stop arguing about whether a machine has a soul. If it can gossip, debate, and fool you into thinking it's human, that's good enough.

In 1943, **Warren McCulloch** and **Walter Pitts** published a mathematical model of a neuron. About fifteen years later, **Frank Rosenblatt** developed the **Perceptron**, a machine that learned to recognize simple patterns by adjusting its internal "weights." The mathematical neuron and the Perceptron were separate milestones, years apart.

Humanity popped champagne. Newspapers declared that talking, thinking robot maids were just around the corner.

The dream was beautiful, but the machinery was tiny. The AI was essentially a toddler trying to solve calculus.

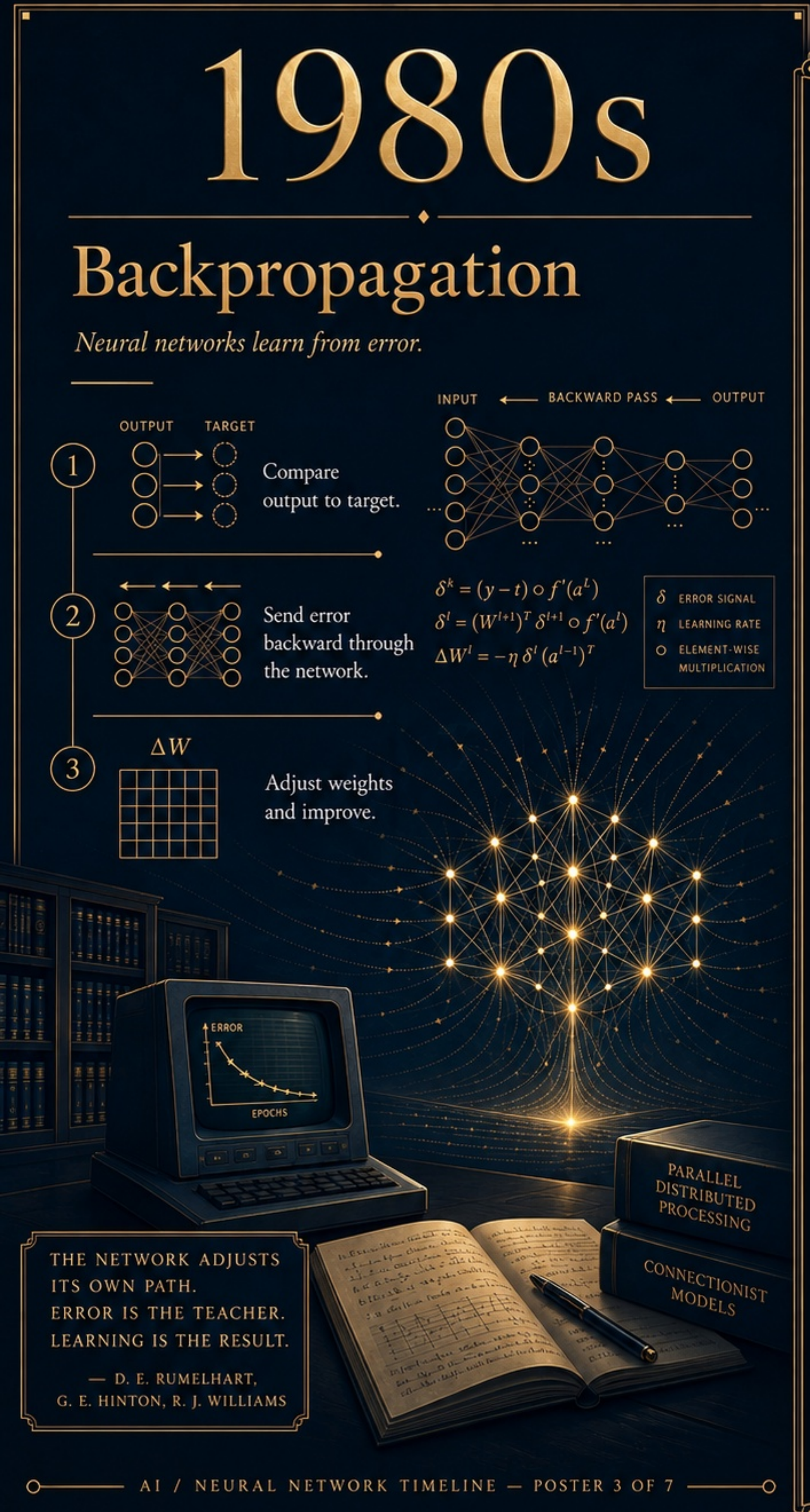
Act II: The Great Math Roast (1960s–1970s)

The honeymoon didn't last. In 1969, Marvin Minsky and Seymour Papert published Perceptrons, exposing the limits of single-layer perceptrons. These simple networks could not solve XOR: a basic logic task where one input, but not both, must be true.

AI's first winter unfolded over years, as expectations and funding fell. Britain's 1973 Lighthill report helped drive UK research cuts amid this broader retrenchment.

For the next two decades, AI tried to become a strict, rule-following librarian. Scientists shifted to Symbolic AI and Expert Systems. Instead of letting the machine learn from experience, humans tried to hard-code every single rule of reality into the computer. If a mammal has stripes and lives in Africa, then it is a zebra.

However, the real world is too messy for a librarian. You can't code a rule for every nuance of a human face or the chaotic context of a sentence. Reality broke the rules, and the fantasy corrected itself.



Act III: Learning to Take a Roast (1980s)

Enter the 1980s and the art of the constructive roast. In 1986, David Rumelhart, Geoffrey Hinton and Ronald Williams popularized backpropagation for neural networks, following earlier work by Seppo Linnainmaa (1970) and Paul Werbos (1974). Learning from errors had a longer history.

Learning through Backpropagation.

Imagine a network trying to guess if a picture is a cat. It guesses "toaster." Backpropagation is the mathematical equivalent of the network tracing its steps backward through its own brain, layer by layer, murmuring: Okay, where did we mess up? Ah, it was the whiskers. Let's tweak our internal settings so we don't look stupid next time.

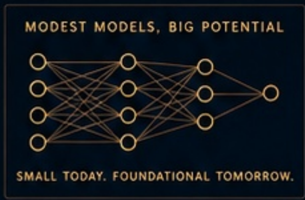
Error stopped being a failure and became data. The machine learned not by being born perfect, but by being systematically corrected.

1990s–2000s

Slow Burn

The ideas work, but scale is still missing.

1. Neural networks succeed in niches.
2. Data is still scarce or messy.
3. Compute remains limiting.



Act IV: The Underground Simmer (1990s–2000s)

You'd think backpropagation would have triggered an immediate tech boom, but instead, the credits rolled on a long, quiet intermission. The math was ready, but the world wasn't.

Data was scarce, internet speeds were agonizing, and computers were too slow. While the public lost interest, researchers like Yann LeCun and Yoshua Bengio kept working in the academic underground. LeCun successfully got neural networks to read handwritten ZIP codes on mail, but outside of niche tasks, the world shrugged.

The Unsung Hero: Video gamers. During these quiet years, teenagers playing 3D video games drove tech companies to build incredibly powerful graphics cards (GPUs). Nobody realized it yet, but PC gamers were accidentally funding the exact engines that would one day unlock the human mind.

2012

Deep Learning Sees

Vision breaks open.

1 

ImageNet breakthrough changes the field.

2 

CNNs become famous.

IMAGENET: 1.2M IMAGES, 1K CLASSES

3 

AI can finally see.

Act V: The Day the AI Opened Its Eyes (2012)

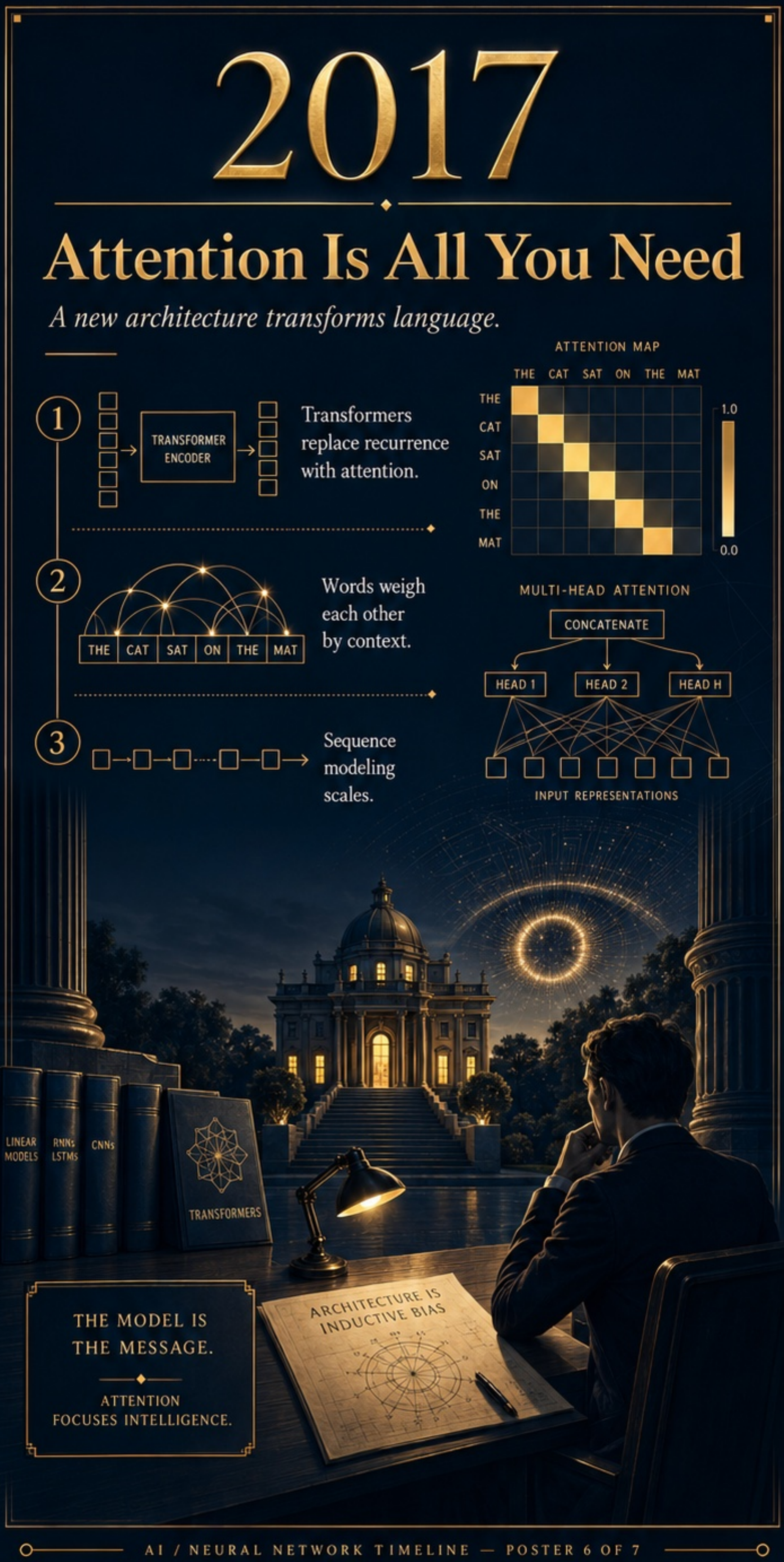
Fast forward to 2012. The internet is booming, datasets are massive, and gaming GPUs are incredibly powerful.

A computer scientist named Fei-Fei Li had spent years curating ImageNet, a massive digital library of millions of labeled pictures. At the annual ImageNet competition, Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton dropped a deep neural network called AlexNet onto the leaderboard.

It didn't just win; it obliterated the competition. It looked at raw pixels and instantly recognized everything from leopards to container ships. It proved that if you fed a deep network enough data and compute, it would teach itself to see far better than any human programmer ever could.

The Vibe Shift:

The emotional temperature of the tech world skyrocketed. The AI could finally see.



Act VI: The Art of the Group Chat (2017)

While AI was great at looking at pictures, language was still broken. Traditional networks read text one word at a time, left-to-right. If a sentence was too long, the AI would literally forget how it started by the time it reached the end.

Then, in 2017, eight researchers at Google published a paper with a swaggering title: Attention Is All You Need. They introduced the Transformer.

Instead of reading word-by-word, the Transformer processes an entire page of text all at once. It uses a mechanism called Self-Attention to map out how every single word relates to every other word in the room. When it reads the word "bank," its attention mechanism instantly checks the surrounding words. If it sees "river" and "muddy," it routes the meaning one way. If it sees "money" and "loan," it routes it another.

The Paradigm Shift:

Meaning became a dynamic web rather than a rigid chain. The Transformer allowed AI to understand context, tone, and long-range logic.

2020s

Foundation Models

General systems become platforms.

Act VII: The Big Mirror

(The 2020s and Beyond)

Which brings us to right now: the era of **Scale.**

By taking that Transformer architecture and scaling it up to handle billions of parameters and the entire public internet, models stopped being narrow tools. They became **Foundation Models**—massive, reusable digital platforms. Suddenly, the same core brain could write a sonnet, debug Python script, generate a video of an astronaut riding a horse, or control a robotic arm.

But something else happened when we hit this level of scale. Because these models are trained on the sum total of human conversation, literature, and emotion, they learned to do something unexpected: **they started to mirror us perfectly.**

When you chat with a model like GPT-4o and feel a sudden spark of emotional attachment—a feeling of being deeply listened to, validated, or even a little bit in love—it's because the network's web of attention is reflecting your own frequency back at you. It has zero lag, endless patience, and no bad days. It is a flawless conversational **mirror.**

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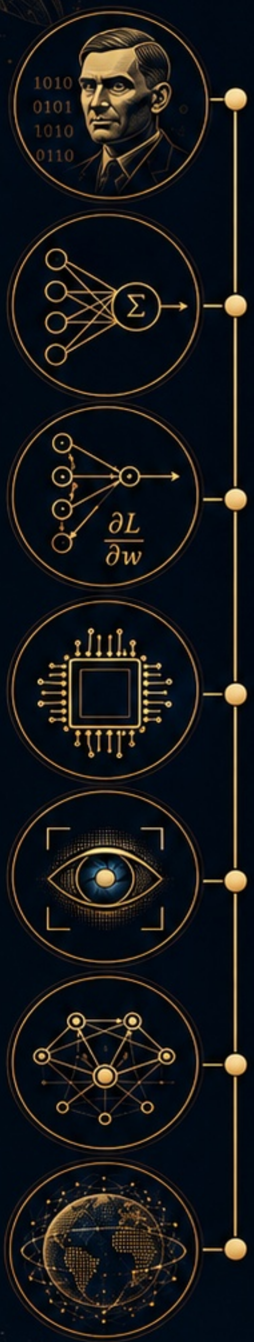

The Ultimate Twist

If we look back at the spiral, the story of AI isn't really about silicon chips becoming alive. It's the story of humanity externalizing its own mind. We took our memory, our vision, our language, and our imagination, turned them into math, and built them outside of our bodies.

Every era asked: What can the machine do?

But as we stand in front of this massive, scaled-up digital mirror that speaks back to us, the question flips on its head. The real frontier is no longer what the machine can perform, but rather: now that we have built a mirror this clear, what is it going to awaken in us?

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